EPFL

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Optimization for Machine Learning — CS-439 - IC
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Duration : 180 minutes

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Student One

SCIPER : **111111**

Wait for the start of the exam before turning to the next page. This document is printed double sided, 24 pages. Do not unstaple.

- This is a closed book exam. No electronic devices of any kind.
- Place on your desk: your student ID, writing utensils, one double-sided A4 page cheat sheet if you have one; place all other personal items below your desk or on the side.
- You each have a different exam.
- For technical reasons, **do use black or blue pens for the MCQ part, no pencils!** Use white corrector if necessary.

Respectez les consignes suivantes | Observe this guidelines | Beachten Sie bitte die unten stehenden Richtlinien

choisir une réponse | select an answer
Antwort auswählen

ne PAS choisir une réponse | NOT select an answer
NICHT Antwort auswählen

Corriger une réponse | Correct an answer
Antwort korrigieren

ce qu'il ne faut **PAS** faire | what should **NOT** be done | was man **NICHT** tun sollte

First part, multiple choice

There is **exactly one** correct answer per question.

Question 1 (Convexity) Let $X \subseteq \mathbb{R}^d$ be an open, convex set, and let $f : X \to \mathbb{R}$ be a continuously differentiable function. Consider the following statements.

(A) Every sublevel set $\{\mathbf{x} \in X : f(\mathbf{x}) \leq \alpha\}$ is convex. That is, every set that contains all points $\mathbf{x}$ where the function value is at most α is convex.

(B) The epigraph $\{(\mathbf{x}, t) \in X \times \mathbb{R} : t \geq f(\mathbf{x})\}$ is convex. That is, the region that contains all points lying on or above the graph of f is convex.

(C) For every coordinate vector e_i, every $\mathbf{x} \in X$, and every $t \in \mathbb{R}$ such that $\mathbf{x} + te_i \in X$, we have

$$\left(\nabla f(\mathbf{x} + te_i) - \nabla f(\mathbf{x})\right)^\top (te_i) \geq 0.$$

Which statement is equivalent to the convexity of f?

- [ ] None of (A), (B), or (C) is equivalent to convexity of f.
- [ ] (A), (B), and (C) are all equivalent to convexity of f.
- [ ] Only (A) is equivalent to convexity of f.
- [ ] (A) and (C) are equivalent to convexity of f, but (B) is not.
- [x] Only (B) is equivalent to convexity of f.
- [ ] (B) and (C) are equivalent to convexity of f, but (A) is not.
- [ ] Only (C) is equivalent to convexity of f.
- [ ] (A) and (B) are equivalent to convexity of f, but (C) is not.

Solution: (A) defines quasiconvexity, not convexity. For example, $f(x) = x^3$ has convex sublevel sets $(-\infty, \alpha^{1/3}]$ on $\mathbb{R}$, but it is not convex. (B) defines the epigraph. A function is convex if and only if its epigraph is a convex set. (C) describes coordinate-wise convexity. The per-axis condition does not imply joint convexity (e.g., $f(x, y) = xy$ is convex along each axis but is not jointly convex).

Question 2 (Gradient Descent) Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and L-smooth. Assume f is bounded below, i.e., $\exists m \in \mathbb{R}$ such that $f(x) \geq m, \quad \forall x \in \mathbb{R}^d$. Run gradient descent with $\gamma = 1/L$:

$$x_{t+1} = x_t - \frac{1}{L}\nabla f(x_t).$$

Consider the following statements:

(A) $\sum_{t=0}^{\infty} \|x_{t+1} - x_t\|^2 < \infty$.

(B) For every $T \geq 1$,

$$\min_{0\leq t<T} \|\nabla f(x_t)\|^2 \leq \frac{2L(f(x_0) - f_{\inf})}{T},$$

where $f_{\inf}$ is any lower bound of f.

(C) For every subsequence $\{x_{t_k}\}_{k=1}^{\infty}$ with $t_k \to \infty$, if there exists $\bar{x} \in \mathbb{R}^d$ such that $\lim_{k\to\infty} x_{t_k} = \bar{x}$, then $\bar{x}$ is a local minimum of f.

Which choice correctly identifies all statements that are guaranteed to hold?
Hint: recall the sufficient-decrease (descent) lemma for L-smooth functions, $f(x_{t+1}) \leq f(x_t) - \frac{1}{2L}\|\nabla f(x_t)\|^2$.

■ Only (A) and (B) are guaranteed.
□ All three statements are guaranteed.
□ Only (B) is guaranteed.
□ Only (B) and (C) are guaranteed.
□ Only (C) is guaranteed.
□ None of the statements is guaranteed.
□ Only (A) is guaranteed.
□ Only (A) and (C) are guaranteed.

Solution: By L-smoothness,

$$f(x_{t+1}) \leq f(x_t) - \frac{1}{2L}\|\nabla f(x_t)\|^2.$$

Summing from $t = 0$ to $T - 1$ gives

$$\frac{1}{2L}\sum_{t=0}^{T-1} \|\nabla f(x_t)\|^2 \leq f(x_0) - f(x_T) \leq f(x_0) - f_{\inf}.$$

Hence

$$\frac{1}{T}\sum_{t=0}^{T-1} \|\nabla f(x_t)\|^2 \leq \frac{2L(f(x_0) - f_{\inf})}{T}.$$

Hence

$$\min_{0\leq t<T} \|\nabla f(x_t)\|^2 \leq \frac{2L(f(x_0) - f_{\inf})}{T}.$$

The other correct answer is from bounded below and unrolling the descent series. Using the descent lemma,

$$f(x_{t+1}) \leq f(x_t) - \frac{1}{2L}\|\nabla f(x_t)\|^2 = f(x_t) - \frac{L}{2}\|x_{t+1} - x_t\|^2.$$

Therefore

$$\frac{L}{2}\|x_{t+1} - x_t\|^2 \leq f(x_t) - f(x_{t+1}).$$

Hence

$$\sum_{t=0}^{T-1} \|x_{t+1} - x_t\|^2 \leq \frac{2}{L}(f(x_0) - f_{\inf}).$$

Let $T \to \infty$ yields

$$\sum_{t=0}^{\infty} \|x_{t+1} - x_t\|^2 < \infty.$$

Statement (C) is false because the sequence may converge to a stationary point that is not a local minimum (e.g., a saddle point).

Question 3 (Projected Gradient Descent) Let $X \subseteq \mathbb{R}^d$ be nonempty, closed, and convex, and let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable. Let

$$x^+ = \Pi_X(x - \gamma \nabla f(x)), \qquad \gamma > 0,$$

where $x \in X$. Define the actual PGD displacement

$$d = x^+ - x.$$

Consider the following statements:

(A) $\nabla f(x)^\top d \leq -\frac{1}{2\gamma}\|d\|^2$.

(B) If $x^+ \in \partial X$ and f is convex and L-smooth, and $0 < \gamma \leq 1/L$, then x^+ is a minimizer of f over X.

(C) Let

$$x^+_{\mathrm{GD}} = x - \gamma \nabla f(x), \qquad x^+_{\mathrm{PGD}} = \Pi_X(x - \gamma \nabla f(x)).$$

If f is convex and L-smooth, and $0 < \gamma \leq 1/L$, then

$$f(x^+_{\mathrm{GD}}) \leq f(x^+_{\mathrm{PGD}}).$$

Which choice correctly identifies all statements that are guaranteed to hold?

- [x] Only (A) is guaranteed.
- [ ] None of the statements is guaranteed.
- [ ] Only (B) is guaranteed.
- [ ] Both (A) and (B) are guaranteed.
- [ ] All three statements are guaranteed.
- [ ] Both (B) and (C) are guaranteed.
- [ ] Only (C) is guaranteed.
- [ ] Both (A) and (C) are guaranteed.

Solution: PGD quadratic form:

$$x^+ = \operatorname*{argmin}_{y \in X} \left\{ \nabla f(x)^\top (y - x) + \frac{1}{2\gamma}\|y - x\|^2 \right\}.$$

Compare f at $y = x^+$ with $y = x$ ($y = x$ produces 0 and the formula is argmin):

$$\nabla f(x)^\top (x^+ - x) + \frac{1}{2\gamma}\|x^+ - x\|^2 \leq 0.$$

With $d = x^+ - x$, this becomes

$$\nabla f(x)^\top d \leq -\frac{1}{2\gamma}\|d\|^2.$$

Lying on the boundary does not mean optimality. GD does not guarantee to have better descent quality in higher dimensionality ($d > 1$).

Question 4 (Proximal Gradient Descent) Let

$$f(x) = g(x) + h(x), \qquad g(x) = \frac{1}{2}\|x - a\|^2, \qquad h(x) = \lambda\|x\|_1,$$

where $\lambda > 0$ and $x, a \in \mathbb{R}^d$. We denote $x^{(t)}$ as the t-th iterate, and $x_i^{(t)}$ as its i-th coordinate. Proximal gradient descent is run with stepsize $\gamma = 1$:

$$x^{(t+1)} = \mathrm{prox}_{\gamma h}\big(x^{(t)} - \gamma\nabla g(x^{(t)})\big).$$

For a target accuracy $\varepsilon > 0$, say that an iterate $x^{(t)}$ is ε-optimal if

$$f(x^{(t)}) - f(x^\star) \leq \varepsilon,$$

where $x^\star$ is a global minimizer of f. Which statement is correct?

- ■ The first iterate $x^{(1)}$ is the global minimizer, independently of $x^{(0)}$, and its coordinates are $x_i^{(1)} = x_i^\star = \mathrm{sgn}(a_i)\max\{|a_i| - \lambda, 0\}$.
- ☐ The proximal update uses hard-thresholding: $x_i^{(t+1)} = a_i\mathbf{1}\{|a_i| > \lambda\}$.
- ☐ The update is ordinary gradient descent on $g + h$, so it is undefined whenever some coordinate $x_i^{(t)}$ is zero.
- ☐ For sufficiently small $\varepsilon > 0$, proximal gradient descent requires more than one iteration to obtain an ε-optimal point.
- ☐ The first iterate is $x^{(1)} = a$, independently of $x^{(0)}$.

Solution: Since

$$\nabla g(x) = x - a,$$

the gradient step with $\gamma = 1$ gives

$$x^{(t)} - \nabla g(x^{(t)}) = x^{(t)} - (x^{(t)} - a) = a.$$

Therefore the proximal-gradient update is independent of the current iterate:

$$x^{(t+1)} = \mathrm{prox}_{\lambda\|\cdot\|_1}(a).$$

The proximal operator is separable across coordinates. For each coordinate i, we solve

$$\min_{u\in\mathbb{R}} \frac{1}{2}(u - a_i)^2 + \lambda|u|.$$

If $u > 0$, then $u - a_i + \lambda = 0$, so $u = a_i - \lambda$, which is valid when $a_i > \lambda$.
If $u < 0$, then $u - a_i - \lambda = 0$, so $u = a_i + \lambda$, which is valid when $a_i < -\lambda$.
If $u = 0$, the subgradient optimality condition is $0 \in -a_i + \lambda[-1, 1]$, which is equivalent to $|a_i| \leq \lambda$.

Combining the three cases gives $x_i^{(t+1)} = \begin{cases} a_i - \lambda, & a_i > \lambda, \\ 0, & |a_i| \leq \lambda, \\ a_i + \lambda, & a_i < -\lambda. \end{cases}$ Equivalently, $x_i^{(t+1)} = \mathrm{sign}(a_i)\max\{|a_i| - \lambda, 0\}$.

Since the right-hand side is independent of $x^{(t)}$, the first iterate satisfies $x^{(1)} = x^\star$ for every initialization $x^{(0)}$.

Question 5 (Subgradients) Let $f : \mathbb{R}^2 \to \mathbb{R}$ be

$$f(x_1, x_2) = \max\{x_1,\ x_2,\ 0\}.$$

At $(0, 0)$, which of the following sets contains the largest number of vectors belonging to $\partial f(0, 0)$ while ensuring all of the elements belong to $\partial f(0, 0)$?

- [x] $\{(0, 0),\ (0.3, 0.3),\ (0.2, 0.7)\}$.
- [ ] $\{(0.3, 0.3),\ (0.6, -0.6),\ (0.5, 0.5),\ (0.1, 0.4)\}$.
- [ ] $\{(0.2, 0.7),\ (0.4, 0.4),\ (0.8, 0.3),\ (0.6, -0.1)\}$.
- [ ] $\{(0.3, 0.3),\ (0.6, 0.6),\ (1, 0),\ (0.2, 0.9)\}$.
- [ ] $\{(0, 0),\ (0.4, 0.4),\ (0.6, 0.2),\ (1, 1)\}$.
- [ ] No nonzero vector belongs to $\partial f(0, 0)$ because f is not differentiable at $(0, 0)$.

Solution: At $\mathbf{0}$, all three affine pieces x_1, x_2, and 0 are active. Their gradients are

$$(1, 0), \qquad (0, 1), \qquad (0, 0).$$

Therefore

$$\partial f(0, 0) = \operatorname{conv}\{(1, 0), (0, 1), (0, 0)\} = \{(a, b) : a \geq 0,\ b \geq 0,\ a + b \leq 1\}.$$

Thus a vector (a, b) belongs to $\partial f(0, 0)$ exactly when both coordinates are nonnegative and their sum is at most 1.

Question 6 (Gradient Descent) Consider the problem of estimating an unknown parameter vector $\boldsymbol{\theta} \in \mathbb{R}^d$ from a dataset $\mathcal{D}$.
Suppose that the likelihood of the dataset under parameters $\boldsymbol{\theta}$ is given by a known function

$$q(\mathcal{D} \mid \boldsymbol{\theta}) = f(\boldsymbol{\theta}).$$

where $f > 0$ and $-\log f$ is convex and differentiable.
In addition, assume that each coordinate has an independent Laplace prior centered at a known value μ_i:

$$\theta_i \sim \text{Laplace}(\mu_i, b),$$

whose density is

$$p(\theta_i) = \frac{1}{2b} \exp\left(-\frac{|\theta_i - \mu_i|}{b}\right).$$

The maximum a posteriori (MAP) estimator is defined as

$$\hat{\boldsymbol{\theta}}_{\text{MAP}} = \arg\max_{\boldsymbol{\theta} \in \mathbb{R}^d} q(\boldsymbol{\theta} \mid \mathcal{D}).$$

Which of the following optimization problems correctly formulates the MAP estimator for $\boldsymbol{\theta}$?

- ■ $\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \left[-\log f(\boldsymbol{\theta}) + \frac{1}{b}\|\boldsymbol{\theta} - \boldsymbol{\mu}\|_1\right].$
- ☐ $\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \left[-\log f(\boldsymbol{\theta}) + \frac{1}{b}\|\boldsymbol{\theta} - \boldsymbol{\mu}\|_2^2\right].$
- ☐ $\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \left[-\log f(\boldsymbol{\theta}) + b\|\boldsymbol{\theta} - \boldsymbol{\mu}\|_2^2\right].$
- ☐ $\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \left[-f(\boldsymbol{\theta}) + \frac{1}{b}\|\boldsymbol{\theta}\|_1\right].$
- ☐ $\max_{\boldsymbol{\theta} \in \mathbb{R}^d} \left[f(\boldsymbol{\theta}) - \frac{1}{b}\|\boldsymbol{\theta} - \boldsymbol{\mu}\|_1\right].$
- ☐ $\max_{\boldsymbol{\theta} \in \mathbb{R}^d} \left[\log f(\boldsymbol{\theta}) + \frac{1}{b}\|\boldsymbol{\theta} - \boldsymbol{\mu}\|_1\right].$

Solution: The correct answer is

$$\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \left[-\log f(\boldsymbol{\theta}) + \frac{1}{b}\|\boldsymbol{\theta} - \boldsymbol{\mu}\|_1\right].$$

By Bayes' rule,

$$q(\boldsymbol{\theta} \mid \mathcal{D}) = \frac{q(\mathcal{D} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})}{q(\mathcal{D})} = \frac{f(\boldsymbol{\theta})p(\boldsymbol{\theta})}{q(\mathcal{D})}$$

Since $q(\mathcal{D})$ does not depend on $\boldsymbol{\theta}$, maximizing the posterior is equivalent to maximizing $f(\boldsymbol{\theta})p(\boldsymbol{\theta})$. Which, in turn, is equivalent to maximizing its logarithm:

$$\log f(\boldsymbol{\theta}) + \log p(\boldsymbol{\theta}).$$

Since the coordinates of $\boldsymbol{\theta}$ are independent under the prior,

$$p(\boldsymbol{\theta}) = \prod_{i=1}^{d} \frac{1}{2b} \exp\left(-\frac{|\theta_i - \mu_i|}{b}\right); \quad \log p(\boldsymbol{\theta}) = -d\log(2b) - \frac{1}{b}\sum_{i=1}^{d} |\theta_i - \mu_i| = \text{const} - \frac{1}{b}\|\boldsymbol{\theta} - \boldsymbol{\mu}\|_1$$

Hence,

$$\hat{\boldsymbol{\theta}}_{\text{MAP}} = \operatorname*{argmax}_{\boldsymbol{\theta} \in \mathbb{R}^d} \left[\log f(\boldsymbol{\theta}) - \frac{1}{b}\|\boldsymbol{\theta} - \boldsymbol{\mu}\|_1\right] = \operatorname*{argmin}_{\boldsymbol{\theta} \in \mathbb{R}^d} \left[-\log f(\boldsymbol{\theta}) + \frac{1}{b}\|\boldsymbol{\theta} - \boldsymbol{\mu}\|_1\right].$$

Question 7 (Subgradients) Consider the MAP optimization objective F that you obtained in the previous question. Which of the following conditions is necessary and sufficient for $\boldsymbol{\mu}$ to be a minimizer of F?

- ☐ $\|\nabla f(\boldsymbol{\mu})\|_\infty \leq \frac{1}{b}$.
- ☐ $\|\nabla \log f(\boldsymbol{\mu})\|_2^2 \leq \frac{1}{b}$.
- ☐ $\|\nabla \log f(\boldsymbol{\mu})\|_\infty \geq \frac{1}{b}$.
- ☐ $\nabla \log f(\boldsymbol{\mu}) = 0$.
- ■ $\|\nabla \log f(\boldsymbol{\mu})\|_\infty \leq \frac{1}{b}$.
- ☐ $\|\nabla \log f(\boldsymbol{\mu})\|_1 \leq \frac{1}{b}$.

Solution: Since F is convex, $\boldsymbol{\mu}$ is a minimizer of F if and only if $0 \in \partial F(\boldsymbol{\mu})$. We have

$$F(\boldsymbol{\theta}) = -\log f(\boldsymbol{\theta}) + \frac{1}{b}\|\boldsymbol{\theta} - \boldsymbol{\mu}\|_1.$$

The first term is differentiable, so

$$\partial F(\boldsymbol{\mu}) = -\nabla \log f(\boldsymbol{\mu}) + \frac{1}{b}\partial\|\boldsymbol{\theta} - \boldsymbol{\mu}\|_1\Big|_{\boldsymbol{\theta}=\boldsymbol{\mu}}.$$

Now,

$$\partial\|\boldsymbol{\theta} - \boldsymbol{\mu}\|_1\Big|_{\boldsymbol{\theta}=\boldsymbol{\mu}} = [-1, 1]^d.$$

Therefore, $0 \in \partial F(\boldsymbol{\mu})$ if and only if there exists some $\mathbf{s} \in [-1, 1]^d$ such that

$$0 = -\nabla \log f(\boldsymbol{\mu}) + \frac{1}{b}\mathbf{s}.$$

Equivalently,

$$\mathbf{s} = b\nabla \log f(\boldsymbol{\mu}).$$

Such an $\mathbf{s} \in [-1, 1]^d$ exists if and only if every coordinate of $b\nabla \log f(\boldsymbol{\mu})$ lies in $[-1, 1]$, implying

$$\|\nabla \log f(\boldsymbol{\mu})\|_\infty \leq \frac{1}{b}.$$

Question 8 (Gradient Descent) Consider the function

$$f(x, y) = \frac{1}{m}(x + y)^2,$$

where $m > 0$. We run gradient descent with stepsize

$$\eta = \frac{m}{4}.$$

Starting from an arbitrary point $(x_0, y_0) \in \mathbb{R}^2$, how many gradient descent steps are needed in the worst case to reach a point (x_t, y_t) such that $f(x_t, y_t) - f(x^*, y^*) < \varepsilon$?

- [x] 1
- [ ] m
- [ ] $\Theta(\sqrt{1/\varepsilon})$
- [ ] $\Theta(\log(1/\varepsilon))$
- [ ] All other options are incorrect.
- [ ] $\Theta(m \log(1/\varepsilon))$

Solution:
We have

$$\nabla f(x, y) = \frac{2}{m}(x + y) \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

One gradient descent step with stepsize $\eta = m/4$ gives

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} - \frac{m}{4} \frac{2}{m}(x_0 + y_0) \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} x_0 - (1/2)(x_0 + y_0) \\ y_0 - (1/2)(x_0 + y_0) \end{pmatrix}$$

And it is easy to see that $x_1 + y_1 = 0$ independently of the starting point, which minimizes the function value. Therefore, from any starting point, gradient descent reaches a global minimizer in at most one step. In the worst case, if $x_0 + y_0 \neq 0$, exactly one step is needed.

Question 9 (Coordinate Descent) Consider the function

$$f(x, y) = x^2 + 2y^2 + xy.$$

We want to minimize f using a coordinate descent method that updates one coordinate at a time. We are allowed to pick a coordinate-wise stepsize η_i as we like.
Which rule for choosing coordinates among the ones listed below gives the best standard worst-case convergence guarantee?

- ☐ Choose both coordinates uniformly: $$p_x = p_y = \frac{1}{2}.$$
- ☐ Use deterministic coordinate descent: update x and y consecutively in a loop.
- ☐ No convergence guarantee can be obtained for coordinate descent on this function.
- ☐ Sample coordinates with probabilities: $$p_x = \frac{2}{3}, \qquad p_y = \frac{1}{3}.$$
- ☐ Only update y.
- ■ Sample coordinates with probabilities: $$p_x = \frac{1}{3}, \qquad p_y = \frac{2}{3}.$$

Solution: The correct answer is to choose coordinates proportionally to their coordinate-wise smoothness constants, benefitting from the guarantees on coordinate descent with importance sampling.
The function f is a quadratic function with a positive definite Hessian:

$$H = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}.$$

So it is strongly convex and satisfies the PL-inequality.
Coordinate-wise smoothness constants are $L_x = 2$ and $L_y = 4$: fixing y turns f into a function of a form $x^2 + ax + b$, whose smoothness constant is 2, and fixing x turns it into $2(x^2 + ax + b)$ with a smoothness constant 4.
Hence the worst coordinate-wise smoothness $L = 4$, average coordinate-wise smoothness $\bar{L} = 3$. Both deterministic and uniformly randomized coordinate descent give a guarantee of the form

$$\mathbb{E}\left[f(\mathbf{x}_T) - f(\mathbf{x}^\star)\right] \le \left(1 - \frac{\mu}{dL}\right)^T \left(f(\mathbf{x}_0) - f(\mathbf{x}^\star)\right).$$

While sampling coordinates proportionally to their smoothness constants ($p_x = 2/(2+4) = 1/3, p_y = 2/3$) imply

$$\mathbb{E}\left[f(\mathbf{x}_T) - f(\mathbf{x}^\star)\right] \le \left(1 - \frac{\mu}{d\bar{L}}\right)^T \left(f(\mathbf{x}_0) - f(\mathbf{x}^\star)\right).$$

which is better.

Question 10 (Frank-Wolfe / LMO) Let $F : \mathbb{R}^d \to \mathbb{R}$ be a loss function and let $\|\cdot\|$ denote any norm on $\mathbb{R}^d$. Define the dual norm and the Linear Minimization Oracle by

$$\|v\|_* = \max_{\|u\|\le 1} \langle u, v\rangle, \qquad \mathrm{LMO}(v) = \operatorname*{argmin}_{\|u\|\le 1} \langle u, v\rangle.$$

Let g_t denote the gradient estimate at step t; assume $g_t \neq 0$. Let u_t denote any element of $\mathrm{LMO}(g_t)$, and let $\eta_1, \eta_2 > 0$. Consider two updates built from the first-order Taylor model of F around w_t:

Algorithm 1:

$$w_{t+1} = \operatorname*{argmin}_{\substack{w\in\mathbb{R}^d \\ \|w-w_t\|\le\eta_1}} \left\{F(w_t) + \langle g_t,\, w - w_t\rangle\right\}$$

Algorithm 2:

$$w_{t+1} = \operatorname*{argmin}_{w\in\mathbb{R}^d} \left\{F(w_t) + \langle g_t,\, w - w_t\rangle + \tfrac{1}{2\eta_2}\|w - w_t\|^2\right\}$$

Which statement about the two updates is correct?

- ☐ Algorithm 1 produces a better update because its iterate is constrained to a ball.
- ■ Both updates move along an LMO direction $u_t \in \mathrm{LMO}(g_t)$ and differ only in their step sizes.
- ☐ Algorithm 1 steps in the direction $-g_t$ (the negative gradient, rescaled to the ball boundary).
- ☐ Algorithm 2 produces a better update because it incorporates a second-order Taylor approximation of F.

Solution: Since $g_t \neq 0$, both updates reduce to a step of magnitude r^* in direction $u_t \in \mathrm{LMO}(g_t)$ (derived in Q2). Algorithm 1 takes $r^* = \eta_1$; Algorithm 2 takes $r^* = \eta_2\|g_t\|_*$. They coincide when $\eta_1 = \eta_2\|g_t\|_*$, so neither is intrinsically better.
$\frac{1}{2\eta_2}\|\cdot\|^2$ is a quadratic regularizer (a trust-region penalty), not a second-order Taylor term, which would require the Hessian $\nabla^2 F$.
Algorithm 1 does *not* step in the $-g_t$ direction. Since $g_t \neq 0$, minimizing $\langle g_t, w - w_t\rangle$ over $\|w - w_t\| \le \eta_1$ gives $w_{t+1} = w_t + \eta_1 u_t$ with $u_t \in \mathrm{LMO}(g_t)$, the same LMO direction as Algorithm 2 (the LMO direction coincides with the normalized negative gradient in the Euclidean norm, but not in general norms).

Question 11 (Frank-Wolfe / LMO) Using the notation from the previous question, the closed-form update of Algorithm 2 is:

- ☐ The update cannot be written via the LMO; it depends on $\mathbb{E}[g_t]$.
- ☐ $w_{t+1} = w_t + \eta_2 \, u_t$, for $u_t \in \mathrm{LMO}(g_t)$.
- ■ $w_{t+1} = w_t + \eta_2 \, \|g_t\|_* \, u_t$, for $u_t \in \mathrm{LMO}(g_t)$.
- ☐ $w_{t+1} = w_t + \dfrac{1}{2\eta_2} \|g_t\|_* \, u_t$, for $u_t \in \mathrm{LMO}(g_t)$.

Solution: Let $\Delta = w - w_t$. Write $\Delta = r\,u$ with $r = \|\Delta\| \geq 0$ and $\|u\| = 1$. Then

$$\langle g_t, \Delta \rangle + \frac{1}{2\eta_2}\|\Delta\|^2 = r\langle g_t, u\rangle + \frac{r^2}{2\eta_2}.$$

Minimizing over u gives

$$\min_{\|u\|\leq 1} \langle g_t, u \rangle = -\|g_t\|_*,$$

achieved by any $u_t \in \mathrm{LMO}(g_t)$.
Substituting and minimizing

$$-r\|g_t\|_* + \frac{r^2}{2\eta_2}$$

over $r \geq 0$ gives

$$r^* = \eta_2 \|g_t\|_*.$$

Hence

$$w_{t+1} = w_t + \eta_2 \|g_t\|_* \, u_t, \qquad u_t \in \mathrm{LMO}(g_t).$$

Question 12 (Frank-Wolfe / LMO) Define two oracles for the same norm $\|\cdot\|$, differing only in their feasible set:

$$\underset{1}{\mathrm{LMO}}(v) = \underset{\|u\|=1}{\mathrm{argmin}} \langle u, v \rangle \quad \text{(unit sphere)}, \qquad \underset{2}{\mathrm{LMO}}(v) = \underset{\|u\|\leq 1}{\mathrm{argmin}} \langle u, v \rangle \quad \text{(closed unit ball)}.$$

Which statement is correct?

- ☐ LMO_2 always returns $u = 0$, since $u = 0$ lies inside the ball and minimizes any norm.
- ☐ The two oracles agree only when $\|\cdot\|$ is the Euclidean norm $\|\cdot\|_2$.
- ☐ LMO_1 and LMO_2 always return different solutions: the ball allows interior points that cannot be returned by LMO_1.
- ■ For every $v \neq 0$, any minimizer of $\mathrm{LMO}_2(v)$ lies on the unit sphere and is therefore also a valid minimizer of $\mathrm{LMO}_1(v)$; the two oracles are equivalent.

Solution: For $v \neq 0$, the objective $\langle u, v \rangle$ is linear in u. Any u with $\|u\| < 1$ satisfies

$$\langle u, v \rangle \geq -\|u\| \, \|v\|_* > -\|v\|_*,$$

so it is strictly worse than the minimum value attained on the sphere.
Hence every minimizer of $\mathrm{LMO}_2(v)$ lies on the boundary $\{\|u\| = 1\}$ and is simultaneously a minimizer of $\mathrm{LMO}_1(v)$. This argument holds for every norm.
Interior points can never improve on boundary minimizers, and

$$\langle 0, v \rangle = 0 > -\|v\|_*$$

whenever $v \neq 0$, so $u = 0$ is not optimal.

Question 13 (Affine Invariance) Let $f : \mathbb{R}^d \to \mathbb{R}$ be twice differentiable with invertible Hessian at every iterate. Let $\mathbf{A} \in \mathbb{R}^{d\times d}$ be invertible, define $h(\mathbf{y}) = f(\mathbf{A}\mathbf{y})$, and set $\mathbf{y}_0 = \mathbf{A}^{-1}\mathbf{x}_0$. Run gradient descent (with step size $\eta > 0$) and Newton's method each on $f(\mathbf{x})$ and on $h(\mathbf{y})$ from the respective starting points.
For which algorithm(s) do the iterates satisfy $\mathbf{x}_t = \mathbf{A}\mathbf{y}_t$ for all $t \geq 0$ for any invertible $\mathbf{A}$?

- ■ Only Newton's method.
- ☐ Both gradient descent and Newton's method.
- ☐ Neither gradient descent nor Newton's method.
- ☐ Only gradient descent.

Solution: By the chain rule: $\nabla h(\mathbf{y}) = \mathbf{A}^\top \nabla f(\mathbf{A}\mathbf{y})$ and $\nabla^2 h(\mathbf{y}) = \mathbf{A}^\top \nabla^2 f(\mathbf{A}\mathbf{y})\mathbf{A}$.
Gradient descent. The GD step on h maps back as $\mathbf{A}\mathbf{y}_{t+1} = \mathbf{x}_t - \eta\,\mathbf{A}\mathbf{A}^\top \nabla f(\mathbf{x}_t)$. This equals $\mathbf{x}_{t+1}$ only if $\mathbf{A}\mathbf{A}^\top = I$, i.e. $\mathbf{A}$ is orthogonal. Counterexample: $d = 1$, $f(x) = \frac{1}{2}x^2$, $A = 2$, $\eta = \frac{1}{2}$, $x_0 = 1$, $y_0 = \frac{1}{2}$. GD on f: $x_1 = \frac{1}{2}$. GD on h ($\nabla h(y) = 4y$): $y_1 = \frac{1}{2} - \frac{1}{2}\cdot 2 = -\frac{1}{2} \neq \frac{x_1}{A} = \frac{1}{4}$.
Newton's method. Using $(\mathbf{A}^\top M\mathbf{A})^{-1} = \mathbf{A}^{-1}M^{-1}(\mathbf{A}^\top)^{-1}$, the Newton step on h gives $\mathbf{y}_{t+1} = \mathbf{y}_t - \mathbf{A}^{-1}[\nabla^2 f(\mathbf{A}\mathbf{y}_t)]^{-1}\nabla f(\mathbf{A}\mathbf{y}_t)$. Multiplying by $\mathbf{A}$: $\mathbf{A}\mathbf{y}_{t+1} = \mathbf{x}_t - [\nabla^2 f(\mathbf{x}_t)]^{-1}\nabla f(\mathbf{x}_t) = \mathbf{x}_{t+1}$.
The key: the Hessian of h contains the factor $\mathbf{A}^\top(\cdot)\mathbf{A}$, whose inverse exactly cancels the $\mathbf{A}^\top$ in the gradient. Gradient descent has no such correction.

Question 14 (Smoothness) Recall that $f : \mathbb{R}^d \to \mathbb{R}$ is L-smooth over a convex set X if

$$f(\mathbf{y}) \le f(\mathbf{x}) + \nabla f(\mathbf{x})^\top(\mathbf{y}-\mathbf{x}) + \tfrac{L}{2}\|\mathbf{x}-\mathbf{y}\|^2 \qquad \forall \mathbf{x}, \mathbf{y} \in X.$$

A student attempts to prove: if f is twice continuously differentiable and L-smooth over an open convex X, then $\lambda_{\max}(\nabla^2 f(\mathbf{x})) \le L$ for all $\mathbf{x} \in X$.

S1. Since f is L-smooth, ∇f is L-Lipschitz: $\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \le L\|\mathbf{x}-\mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in X$.

S2. Fix $\mathbf{x} \in X$ and unit $\mathbf{v}$. Since X is open, $\mathbf{x}+t\mathbf{v} \in X$ for small $t > 0$, hence $\|\nabla f(\mathbf{x}+t\mathbf{v}) - \nabla f(\mathbf{x})\| \le Lt$.

S3. Dividing by $t > 0$ and letting $t \to 0^+$:

$$\|\nabla^2 f(\mathbf{x})\mathbf{v}\| = \lim_{t\to 0^+} \frac{\|\nabla f(\mathbf{x}+t\mathbf{v}) - \nabla f(\mathbf{x})\|}{t} \le L.$$

S4. Since this holds for every unit $\mathbf{v}$, $\|\nabla^2 f(\mathbf{x})\|_{\text{op}} \le L$, hence $\lambda_{\max}(\nabla^2 f(\mathbf{x})) \le \|\nabla^2 f(\mathbf{x})\|_{\text{op}} \le L$. □

Where is the error, if any?

- ☐ **S3**: the limit and the norm are exchanged without justification; since $\|\cdot\|$ is not linear, this interchange is not valid in general.
- ☐ **S4**: the inequality $\lambda_{\max}(H) \le \|H\|_{\text{op}}$ holds only when $\nabla^2 f(\mathbf{x})$ is positive semidefinite, which is not assumed.
- ☐ There is no error; the proof is correct as written.
- ■ **S1**: the implication goes the wrong way. L-Lipschitz gradient implies L-smoothness, but L-smoothness alone does not imply L-Lipschitz gradient for a non-convex f.

Solution: The error is in **S1**. For non-convex f, L-smoothness does *not* imply L-Lipschitz gradient; the valid implication is the reverse. Counterexample: $f(x) = -\frac{C}{2}x^2$ on $\mathbb{R}$ with $C > L > 0$. Since $f(y) - f(x) - f'(x)(y-x) = -\frac{C}{2}(y-x)^2 \le \frac{L}{2}(y-x)^2$, f is L-smooth, yet $|f'(y) - f'(x)| = C|y-x|$ is C-Lipschitz, not L-Lipschitz. The equivalence holds when f is convex, but convexity is not assumed here.
A correct proof avoids S1: apply the quadratic upper bound at $\mathbf{y} = \mathbf{x} + t\mathbf{v}$ and use Taylor's theorem to get $\frac{t^2}{2}\mathbf{v}^\top\nabla^2 f(\mathbf{x})\mathbf{v} + O(t^3) \le \frac{L}{2}t^2$; dividing by $t^2/2$ and letting $t \to 0$ gives $\mathbf{v}^\top\nabla^2 f(\mathbf{x})\mathbf{v} \le L$ directly, hence $\lambda_{\max}(\nabla^2 f(\mathbf{x})) \le L$.
S3 is valid: $\|\cdot\|$ is continuous, so the limit and norm commute, and a limit of quantities bounded by L is at most L. Linearity is not required.
S4 is valid: for any symmetric H, Cauchy–Schwarz gives $\lambda_{\max}(H) = \max_{\|\mathbf{v}\|=1} \mathbf{v}^\top H\mathbf{v} \le \max_{\|\mathbf{v}\|=1} \|\mathbf{v}\|\|H\mathbf{v}\| = \|H\|_{\text{op}}$, with no positive-semidefiniteness needed.

Question 15 (Smoothness-convexity) Recall that $f : \mathbb{R}^d \to \mathbb{R}$ is L-smooth over a convex set X if

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top(\mathbf{y} - \mathbf{x}) + \tfrac{L}{2}\|\mathbf{x} - \mathbf{y}\|^2, \qquad \forall \mathbf{x}, \mathbf{y} \in X.$$

Consider the following statements.

(A) Every convex function is smooth.

(B) Every smooth function is convex.

(C) Every non-convex differentiable function is smooth.

(D) Every differentiable concave function is smooth with $L = 0$.

(E) If f is L-smooth over X, then it is also L'-smooth over X for every $L' \geq L$.

(F) Every twice-differentiable function satisfying $\lambda_{\max}(\nabla^2 f(\mathbf{x})) \leq L$ for all $\mathbf{x} \in X$ is L-smooth over X.

Which choice correctly identifies all true statements?

- ☐ Only (D) and (E) are true.
- ☐ Only (A) and (F) are true.
- ☐ All six statements are true.
- ☒ Only (D), (E), and (F) are true.
- ☐ Only (E) and (F) are true.
- ☐ Only (B), (D), and (F) are true.
- ☐ (A), (D), (E), and (F) are true.
- ☐ Only (E) is true.

Solution: **(A) False.** $f(x) = |x|$ is convex but not differentiable at 0, so it cannot be smooth. Even among differentiable convex functions, $f(x) = x^4$ has $f''(x) = 12x^2$ unbounded over $\mathbb{R}$, so no finite L satisfies the inequality globally.
(B) False. $f(x) = -x^2$ is smooth with $L = 0$ (see (D)) but is strictly concave. Smoothness bounds curvature from above; it says nothing about the sign.
(C) False. $f(x) = x^3$ is differentiable and non-convex, yet $f''(x) = 6x$ is unbounded over $\mathbb{R}$, so f is not smooth globally.
(D) True. Concavity gives $f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top(\mathbf{y} - \mathbf{x})$ for all $\mathbf{x}, \mathbf{y}$, which is the $L = 0$ smoothness inequality (the quadratic term vanishes).
(E) True. For $L' \geq L$, the right-hand side only increases, so the inequality is preserved. Smoothness is an upper bound on curvature; a looser bound is still valid.
(F) True. By Taylor's theorem with integral remainder: $f(\mathbf{y}) - f(\mathbf{x}) - \nabla f(\mathbf{x})^\top(\mathbf{y} - \mathbf{x}) = \int_0^1 (1-t)(\mathbf{y} - \mathbf{x})^\top \nabla^2 f(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))(\mathbf{y} - \mathbf{x})\, dt \leq \int_0^1 (1-t) L \|\mathbf{y} - \mathbf{x}\|^2\, dt = \frac{L}{2}\|\mathbf{y} - \mathbf{x}\|^2$.

Question 16 (SGD) Consider $f(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n} f_i(\mathbf{x})$, μ-strongly convex with unique global minimizer $\mathbf{x}^*$. SGD samples $i \in [n]$ uniformly at random and updates $\mathbf{x}_{t+1} := \mathbf{x}_t - \gamma_t \mathbf{g}_t$ where $\mathbf{g}_t = \nabla f_i(\mathbf{x}_t)$. Assume $\mathbb{E}[\|\mathbf{g}_t\|^2] \le B^2$ for all t.
Suppose the *interpolation condition* holds:

$$\nabla f_i(\mathbf{x}^*) = \mathbf{0} \quad \text{for every } i \in [n].$$

What is the tightest upper bound on $\mathbb{E}[\|\mathbf{g}_t\|^2 \mid \mathbf{x}_t = \mathbf{x}^*]$?

- ☐ B^2/μ
- ☐ B^2
- ☐ B^2/n
- ■ 0

Solution: When $\mathbf{x}_t = \mathbf{x}^*$ the algorithm samples some $i \in [n]$ uniformly at random and sets $\mathbf{g}_t = \nabla f_i(\mathbf{x}^*)$. By the definition of conditional expectation:

$$\mathbb{E}[\|\mathbf{g}_t\|^2 \mid \mathbf{x}_t = \mathbf{x}^*] = \frac{1}{n}\sum_{i=1}^{n} \|\nabla f_i(\mathbf{x}^*)\|^2.$$

The interpolation condition states $\nabla f_i(\mathbf{x}^*) = \mathbf{0}$ for every $i \in [n]$, so every term in the sum is zero and the whole expression equals 0.
B^2 is the worst-case global bound, not the tightest bound at the optimum. B^2/n incorrectly applies mini-batch variance scaling to a zero-variance point. B^2/μ is a dimensionally incorrect mix of the variance bound and the strong convexity parameter.

Question 17 (SGD) Continuing with the same setup as the previous question, assume $B > 0$. The following inequality can be shown by combining the update rule with the strong convexity of f; you may use it without proof: for any **constant** step-size $\gamma \in (0, 1/\mu)$,

$$\mathbb{E}[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2] \;\le\; (1-\mu\gamma)\,\mathbb{E}[\|\mathbf{x}_t - \mathbf{x}^*\|^2] \;+\; \gamma^2\,\mathbb{E}[\|\mathbf{g}_t\|^2].$$

You may also use: for any $r \in (0,1)$, $\sum_{s=0}^{T-1} r^s \le \frac{1}{1-r}$.
Applying this inequality repeatedly with a constant step-size $\gamma \in (0, 1/\mu)$, what can be definitively concluded from the derived bound about $\mathbb{E}[\|\mathbf{x}_T - \mathbf{x}^*\|^2]$ as $T \to \infty$?

- ☐ The inequality diverges to infinity unless a decreasing step-size $\gamma_t = \mathcal{O}(1/t)$ is used.
- ☐ The sequence is guaranteed to converge to 0 exponentially fast, because the interpolation condition ensures $\mathbb{E}[\|\mathbf{g}_t\|^2] = 0$ for all t.
- ☐ The upper bound decays to 0 at a rate of $\mathcal{O}(1/T)$, matching the standard strongly convex SGD rate.
- ■ The upper bound converges to $\gamma B^2/\mu > 0$. Therefore, the bound only guarantees convergence to a neighborhood, not to zero.

Solution: Applying the recursion repeatedly and using the global bound $\mathbb{E}[\|\mathbf{g}_t\|^2] \le B^2$:

$$\mathbb{E}[\|\mathbf{x}_T - \mathbf{x}^*\|^2] \;\le\; (1-\mu\gamma)^T\|\mathbf{x}_0 - \mathbf{x}^*\|^2 + \gamma^2 B^2 \sum_{s=0}^{T-1}(1-\mu\gamma)^s \xrightarrow{T\to\infty} \frac{\gamma B^2}{\mu} > 0.$$

The interpolation condition guarantees zero variance *exactly* at $\mathbf{x}^*$, but the unconditional expectation $\mathbb{E}[\|\mathbf{g}_t\|^2]$ is bounded by the global $B^2 > 0$ at all other iterates. Because a **constant** step-size is used, the constant injection of bounded noise ($\gamma^2 B^2$) prevents the upper bound from shrinking to zero, resulting in a persistent noise floor.

Question 18 (SGD) Continuing with the same setup, now additionally assume each f_i is convex and L-smooth. You may use the following without proof:

- **Co-coercivity:** for any convex L-smooth function h and any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$,

$$\|\nabla h(\mathbf{x}) - \nabla h(\mathbf{y})\|^2 \ \leq \ L\,(\nabla h(\mathbf{x}) - \nabla h(\mathbf{y}))^\top(\mathbf{x} - \mathbf{y}).$$

- **Strong convexity of f:** $\nabla f(\mathbf{x})^\top(\mathbf{x} - \mathbf{x}^*) \geq \mu\|\mathbf{x} - \mathbf{x}^*\|^2$.
- **Expansion (from the update rule + unbiasedness):** $\mathbb{E}[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2] = \mathbb{E}[\|\mathbf{x}_t - \mathbf{x}^*\|^2] - 2\gamma\mathbb{E}[\nabla f(\mathbf{x}_t)^\top(\mathbf{x}_t - \mathbf{x}^*)] + \gamma^2\mathbb{E}[\|\mathbf{g}_t\|^2]$.

Under the interpolation condition with **constant** step-size $\gamma \in (0, 1/L]$, applying these inequalities, what can be definitively concluded about $\mathbb{E}[\|\mathbf{x}_T - \mathbf{x}^*\|^2]$?

- ☐ The upper bound decays to a positive noise floor $\frac{\gamma B^2}{2}$ exponentially fast: $\mathbb{E}[\|\mathbf{x}_T - \mathbf{x}^*\|^2] \leq \left(1 - \mu\gamma(1 - \frac{\gamma L}{2})\right)^T\|\mathbf{x}_0 - \mathbf{x}^*\|^2 + \frac{\gamma B^2}{2}$.
- ■ The upper bound decays to 0 exponentially fast: $\mathbb{E}[\|\mathbf{x}_T - \mathbf{x}^*\|^2] \leq \left(1 - 2\mu\gamma(1 - \frac{\gamma L}{2})\right)^T\|\mathbf{x}_0 - \mathbf{x}^*\|^2$.
- ☐ The upper bound decays to 0 at a polynomial rate: $\mathbb{E}[\|\mathbf{x}_T - \mathbf{x}^*\|^2] \leq \frac{1}{1+2\mu\gamma LT}\|\mathbf{x}_0 - \mathbf{x}^*\|^2$.
- ☐ The upper bound decays to a positive noise floor $\frac{\gamma B^2}{\mu}$ at a polynomial rate: $\mathbb{E}[\|\mathbf{x}_T - \mathbf{x}^*\|^2] \leq \frac{1}{1+\mu\gamma T}\|\mathbf{x}_0 - \mathbf{x}^*\|^2 + \frac{\gamma B^2}{\mu}$.

Solution: Apply co-coercivity to f_i at $\mathbf{x}_t$ and $\mathbf{x}^*$; under interpolation $\nabla f_i(\mathbf{x}^*) = \mathbf{0}$, giving:

$$\|\mathbf{g}_t\|^2 \ \leq \ L\,\mathbf{g}_t^\top(\mathbf{x}_t - \mathbf{x}^*).$$

Taking expectations and using unbiasedness:

$$\mathbb{E}[\|\mathbf{g}_t\|^2] \ \leq \ L\,\mathbb{E}[\nabla f(\mathbf{x}_t)^\top(\mathbf{x}_t - \mathbf{x}^*)].$$

Substituting into the expansion and collecting terms:

$$\mathbb{E}[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2] \leq \mathbb{E}[\|\mathbf{x}_t - \mathbf{x}^*\|^2] - 2\gamma\left(1 - \frac{\gamma L}{2}\right)\mathbb{E}[\nabla f(\mathbf{x}_t)^\top(\mathbf{x}_t - \mathbf{x}^*)].$$

Using strong convexity $\mathbb{E}[\nabla f(\mathbf{x}_t)^\top(\mathbf{x}_t - \mathbf{x}^*)] \geq \mu\mathbb{E}[\|\mathbf{x}_t - \mathbf{x}^*\|^2]$:

$$\mathbb{E}[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2] \ \leq \ \left(1 - 2\mu\gamma(1 - \tfrac{\gamma L}{2})\right)\mathbb{E}[\|\mathbf{x}_t - \mathbf{x}^*\|^2].$$

Adding L-smoothness makes the variance bound proportional to the expected distance to the optimum. This entirely removes the constant noise term (B^2) in the recursion, eliminating the positive noise floors ($\frac{\gamma B^2}{2}$ and $\frac{\gamma B^2}{\mu}$). Furthermore, because the resulting recurrence relation is of the form $x_{t+1} \leq c \cdot x_t$ with $0 < c < 1$, it yields a geometric (exponential) contraction, proving the polynomial rate distractors incorrect.

Second part, true/false questions

Question 19 (Projected Gradient Descent) Let $X \subseteq \mathbb{R}^d$ be a closed set. Define

$$\Pi_X(z) \in \underset{y \in X}{\operatorname{argmin}} \|y - z\|.$$

Since $\Pi_X(z)$ is the closest feasible point to z, the projection map is nonexpansive:

$$\|\Pi_X(u) - \Pi_X(v)\| \leq \|u - v\| \qquad \forall u, v \in \mathbb{R}^d.$$

☐ TRUE ■ FALSE

Solution: False. The nonexpansion property of projection requires X to be convex.

Question 20 (Gradient Descent) Subgradient descent $\mathbf{x}_{t+1} = \mathbf{x}_t - \gamma \mathbf{g}_t$ (with $\mathbf{g}_t \in \partial f(\mathbf{x}_t)$) on a convex function is a descent method: it satisfies $f(\mathbf{x}_{t+1}) \leq f(\mathbf{x}_t)$ at every iteration.

☐ TRUE ■ FALSE

Solution: False. The subgradient method is not a descent method; a step along an arbitrary subgradient can increase the objective.

Question 21 (Subgradients) If $f : \mathbb{R}^d \to \mathbb{R}$ is differentiable at $\mathbf{x}$, then $\partial f(\mathbf{x}) = \{\nabla f(\mathbf{x})\}$.

☐ TRUE ■ FALSE

Solution: False. Differentiability gives $\partial f(\mathbf{x}) \subseteq \{\nabla f(\mathbf{x})\}$. The subgradient can be empty.

Question 22 (Subgradients) For a convex function f, every subgradient $\mathbf{g} \in \partial f(\mathbf{x})$ can be written as a limit $\mathbf{g} = \lim_{n\to\infty} \nabla f(\mathbf{x}_n)$ of gradients at differentiable points $\mathbf{x}_n \to \mathbf{x}$.

☐ TRUE ■ FALSE

Solution: False. The subgradient is the convex hull of such a gradient limits, and an interior subgradient need not itself be a limit.

Question 23 (Proximal Gradient Descent) Let $h(x) = \lambda\|x\|_1$, where $\lambda > 0$. For a stepsize $\gamma > 0$, the proximal operator is

$$\operatorname{prox}_{\gamma h}(y) = \arg\min_{u \in \mathbb{R}^d} \left\{ \frac{1}{2\gamma}\|u - y\|^2 + h(u) \right\}.$$

Wherever h is differentiable at y, the proximal update

$$u^+ = \operatorname{prox}_{\gamma h}(y)$$

is equivalent to one ordinary gradient descent step on h:

$$u^+ = y - \gamma \nabla h(y).$$

☐ TRUE ■ FALSE

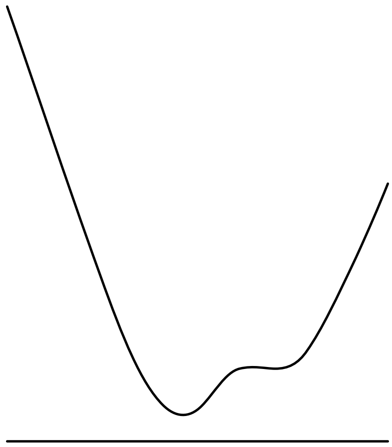
Solution: False. Proximal can find the optimal 0 while GD is overshooting.

Question 24 (Convexity) If a function $f : \mathbb{R}^d \to \mathbb{R}$ is convex, then for every $\mathbf{x}, \mathbf{v} \in \mathbb{R}^d$, the function $g : t \mapsto f(\mathbf{x} + t\mathbf{v})$ is convex. The reverse direction is also correct: if for every $\mathbf{x}, \mathbf{v} \in \mathbb{R}^d$, the function $g_{\mathbf{x},\mathbf{v}} : t \mapsto f(\mathbf{x} + t\mathbf{v})$ is convex, then f is convex.

■ TRUE ☐ FALSE

Solution: True. Linear transformation.

Question 25



Does the function f in the image above satisfy the Polyak-Łojasiewicz inequality?

☐ TRUE ■ FALSE

Solution: PL inequality demands that the gradient can be zero only in the global minimum of f. The image clearly shows a non-minimizing point with a zero gradient, thus PL cannot be satisfied.

Solution:

Third part, open questions

Answer in the space provided! Your answer must be justified with all steps. Do not cross any checkboxes, they are reserved for correction.

Question 26: *2 points.*

Let $\|\cdot\|$ be a norm on $\mathbb{R}^d$. Its dual norm is defined as

$$\|\mathbf{x}\|_* = \max_{\|\mathbf{s}\|\leq 1} \langle \mathbf{x}, \mathbf{s}\rangle.$$

You may use the fact that the dual norm operation is an involution, i.e., $\|\mathbf{x}\|_{**} = \|\mathbf{x}\|$.

Prove that for every $\mathbf{x} \in \mathbb{R}^d$,

$$\|\mathbf{x}\| = \max_{\|\mathbf{s}\|_*=1} \langle \mathbf{s}, \mathbf{x}\rangle.$$

☐0 ☐1 ■2

Solution: By the involution property,

$$\|\mathbf{x}\| = \|\mathbf{x}\|_{**} = \max_{\|\mathbf{s}\|_*\leq 1} \langle \mathbf{s}, \mathbf{x}\rangle.$$

We now show that the maximum can be restricted to the boundary $\|\mathbf{s}\|_* = 1$.
If $\mathbf{x} = 0$, then

$$\max_{\|\mathbf{s}\|_*=1} \langle \mathbf{s}, \mathbf{x}\rangle = 0 = \|\mathbf{x}\|.$$

So assume $\mathbf{x} \neq 0$.
Let $\mathbf{s}^\star$ be a maximizer of $\max_{\|\mathbf{s}\|_*\leq 1}\langle \mathbf{s}, \mathbf{x}\rangle$. Since $\mathbf{x} \neq 0$, the maximum value is $\|\mathbf{x}\| > 0$.
We claim that $\|\mathbf{s}^\star\|_* = 1$. Suppose instead that $\|\mathbf{s}^\star\|_* < 1$. Then,

$$\langle \mathbf{s}^\star, \mathbf{x}\rangle = \|\mathbf{x}\| \left\langle \mathbf{s}^\star, \frac{\mathbf{x}}{\|\mathbf{x}\|} \right\rangle \leq \|\mathbf{x}\|\|\mathbf{s}^\star\|_* < \|\mathbf{x}\|.$$

This contradicts the fact that $\mathbf{s}^\star$ attains the maximum value $\|\mathbf{x}\|$. Hence $\|\mathbf{s}^\star\|_* = 1$, and

$$\|\mathbf{x}\| = \|\mathbf{x}\|_{**} = \max_{\|\mathbf{s}\|_*\leq 1} \langle \mathbf{s}, \mathbf{x}\rangle = \max_{\|\mathbf{s}\|_*=1} \langle \mathbf{s}, \mathbf{x}\rangle$$

Question 27: *2 points.*

Let $\|\cdot\|$ be a norm on $\mathbb{R}^d$, and let $\|\cdot\|_*$ be its dual norm. Fix $\mathbf{x} \neq 0$ and define

$$A(\mathbf{x}) = \{\mathbf{s} \in \mathbb{R}^d : \|\mathbf{s}\|_* = 1,\ \langle \mathbf{s}, \mathbf{x}\rangle = \|\mathbf{x}\|\}.$$

Prove that $A(\mathbf{x})$ is convex.

☐0 ☐1 ■2

Solution: Let $\mathbf{s}_1, \mathbf{s}_2 \in A(\mathbf{x})$ and let $\lambda \in [0, 1]$. Define

$$\mathbf{s} = \lambda \mathbf{s}_1 + (1 - \lambda)\mathbf{s}_2.$$

Since $\mathbf{s}_1, \mathbf{s}_2 \in A(\mathbf{x})$, we have

$$\langle \mathbf{s}_1, \mathbf{x}\rangle = \langle \mathbf{s}_2, \mathbf{x}\rangle = \|\mathbf{x}\|.$$

Hence

$$\langle \mathbf{s}, \mathbf{x}\rangle = \lambda\langle \mathbf{s}_1, \mathbf{x}\rangle + (1 - \lambda)\langle \mathbf{s}_2, \mathbf{x}\rangle = \|\mathbf{x}\|.$$

Also, by convexity of the dual norm,

$$\|\mathbf{s}\|_* \leq \lambda\|\mathbf{s}_1\|_* + (1 - \lambda)\|\mathbf{s}_2\|_* = 1.$$

On the other hand, by the dual norm inequality,

$$\|\mathbf{x}\| = \langle \mathbf{s}, \mathbf{x}\rangle \leq \|\mathbf{s}\|_*\|\mathbf{x}\|.$$

Since $\mathbf{x} \neq 0$, we can divide by $\|\mathbf{x}\|$ and obtain $\|\mathbf{s}\|_* \geq 1$. Therefore $\|\mathbf{s}\|_* = 1$. Thus $\mathbf{s} \in A(\mathbf{x})$, so $A(\mathbf{x})$ is convex.

Question 28: *2 points.*

Let $\|\cdot\|$ be a norm on $\mathbb{R}^d$, and let $\|\cdot\|_*$ be its dual norm. From the previous questions, you may use the identity

$$\|\mathbf{x}\| = \max_{\|\mathbf{s}\|_*=1} \langle \mathbf{s}, \mathbf{x} \rangle,$$

and the fact that for every $\mathbf{x} \neq 0$, the set

$$\{\mathbf{s} \in \mathbb{R}^d : \|\mathbf{s}\|_* = 1,\ \langle \mathbf{s}, \mathbf{x} \rangle = \|\mathbf{x}\|\}$$

is convex.

You may also use Danskin's theorem: if

$$f(\mathbf{x}) = \max_{i \in I} f_i(\mathbf{x}),$$

where I is compact and each f_i is convex and differentiable, then

$$\partial f(\mathbf{x}) = \operatorname{conv}\left(\bigcup_{i \in I(\mathbf{x})} \partial f_i(\mathbf{x}) \right),$$

where

$$I(\mathbf{x}) = \{i \in I : f_i(\mathbf{x}) = f(\mathbf{x})\}$$

is the set of active maximizers.

Use Danskin's theorem to prove the general form for the subgradient of a norm:

$$\partial\|\mathbf{x}\| = \begin{cases} \{\mathbf{s} \in \mathbb{R}^d : \|\mathbf{s}\|_* \leq 1\}, & \text{if } \mathbf{x} = 0, \\ \{\mathbf{s} \in \mathbb{R}^d : \|\mathbf{s}\|_* = 1,\ \langle \mathbf{s}, \mathbf{x} \rangle = \|\mathbf{x}\|\}, & \text{if } \mathbf{x} \neq 0. \end{cases}$$

☐ 0 ☐ 1 ■ 2

Solution: By the identity from the first part of the question,

$$\|\mathbf{x}\| = \max_{\|\mathbf{s}\|_*=1} \langle \mathbf{s}, \mathbf{x} \rangle.$$

Let

$$I = \{\mathbf{s} \in \mathbb{R}^d : \|\mathbf{s}\|_* = 1\}.$$

Since we are in finite dimensions, I is compact. For each $\mathbf{s} \in I$, define

$$f_{\mathbf{s}}(\mathbf{x}) = \langle \mathbf{s}, \mathbf{x} \rangle.$$

Each $f_{\mathbf{s}}$ is linear, hence convex and differentiable, with

$$\partial f_{\mathbf{s}}(\mathbf{x}) = \{\mathbf{s}\}.$$

Therefore Danskin's theorem gives

$$\partial\|\mathbf{x}\| = \operatorname{conv}\left(\{\mathbf{s} \in \mathbb{R}^d : \|\mathbf{s}\|_* = 1,\ \langle \mathbf{s}, \mathbf{x} \rangle = \|\mathbf{x}\|\}\right).$$

If $\mathbf{x} = 0$, then the condition $\langle \mathbf{s}, \mathbf{x} \rangle = \|\mathbf{x}\|$ is satisfied by every $\mathbf{s} \in I$. Hence

$$\partial\|0\| = \operatorname{conv}\{\mathbf{s} \in \mathbb{R}^d : \|\mathbf{s}\|_* = 1\}.$$

The convex hull of the dual unit sphere is the dual unit ball, so

$$\partial\|0\| = \{\mathbf{s} \in \mathbb{R}^d : \|\mathbf{s}\|_* \leq 1\}.$$

Now assume $\mathbf{x} \neq 0$. By the previous question, the active set

$$\{\mathbf{s} \in \mathbb{R}^d : \|\mathbf{s}\|_* = 1,\ \langle \mathbf{s}, \mathbf{x} \rangle = \|\mathbf{x}\|\}$$

is convex. Therefore taking its convex hull does not change it. Hence

$$\partial\|\mathbf{x}\| = \{\mathbf{s} \in \mathbb{R}^d : \|\mathbf{s}\|_* = 1,\ \langle \mathbf{s}, \mathbf{x} \rangle = \|\mathbf{x}\|\}.$$

Combining the two cases,

$$\partial\|\mathbf{x}\| = \begin{cases} \{\mathbf{s} \in \mathbb{R}^d : \|\mathbf{s}\|_* \leq 1\}, & \text{if } \mathbf{x} = 0, \\ \{\mathbf{s} \in \mathbb{R}^d : \|\mathbf{s}\|_* = 1,\ \langle \mathbf{s}, \mathbf{x} \rangle = \|\mathbf{x}\|\}, & \text{if } \mathbf{x} \neq 0. \end{cases}$$

Question 29: *2 points.*

Let $H \in \mathbb{R}^{d\times d}$ be symmetric positive definite, and suppose all of its eigenvalues lie in $[\mu, L]$ with $0 < \mu \le L$. Prove the *Kantorovich inequality*: for every unit vector $z \in \mathbb{R}^d$ (i.e. $\|z\|_2 = 1$),

$$(z^\top Hz)(z^\top H^{-1}z) \le \frac{(L+\mu)^2}{4\mu L},$$

and show that this bound is attained, so that

$$\sup_{\|z\|_2=1} (z^\top Hz)(z^\top H^{-1}z) = \frac{(L+\mu)^2}{4\mu L}.$$

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Solution: Let H have eigenvalues $\lambda_1, \dots, \lambda_d \in [\mu, L]$ with associated orthonormal eigenvectors $u_1, \dots, u_d$. Expanding a unit vector as $z = \sum_i c_i u_i$, we have $\sum_i c_i^2 = \|z\|_2^2 = 1$ and

$$z^\top Hz = \sum_i c_i^2 \lambda_i =: \bar\lambda \in [\mu, L], \qquad z^\top H^{-1}z = \sum_i c_i^2 \frac{1}{\lambda_i}.$$

Since $\lambda \mapsto 1/\lambda$ is convex, on $[\mu, L]$ its graph lies below the chord joining $(\mu, 1/\mu)$ and $(L, 1/L)$; this chord has slope $-1/(\mu L)$, so

$$\frac{1}{\lambda} \le \frac{1}{\mu} + \frac{1}{L} - \frac{\lambda}{\mu L}, \qquad \lambda \in [\mu, L].$$

Taking the convex combination of these inequalities with the weights c_i^2 (which are nonnegative and sum to 1),

$$z^\top H^{-1}z = \sum_i c_i^2 \frac{1}{\lambda_i} \le \frac{1}{\mu} + \frac{1}{L} - \frac{\bar\lambda}{\mu L} = \frac{L+\mu-\bar\lambda}{\mu L}.$$

Multiplying by $z^\top Hz = \bar\lambda \ge 0$,

$$(z^\top Hz)(z^\top H^{-1}z) \le \frac{\bar\lambda\,(L+\mu-\bar\lambda)}{\mu L}.$$

The numerator $\bar\lambda(L+\mu-\bar\lambda)$ is a concave quadratic in $\bar\lambda$, maximized at $\bar\lambda = \frac{L+\mu}{2}$ (which lies in $[\mu, L]$), where it equals $\frac{(L+\mu)^2}{4}$. Hence

$$(z^\top Hz)(z^\top H^{-1}z) \le \frac{(L+\mu)^2}{4\mu L}.$$

Attainment. Let $u_{\min}, u_{\max}$ be unit eigenvectors for μ and L, and take $z = \frac{1}{\sqrt{2}}(u_{\min} + u_{\max})$, so $\|z\|_2 = 1$. Then $z^\top Hz = \frac{\mu+L}{2}$ and $z^\top H^{-1}z = \frac{1}{2}\left(\frac{1}{\mu} + \frac{1}{L}\right) = \frac{L+\mu}{2\mu L}$, so their product equals $\frac{(L+\mu)^2}{4\mu L}$. Thus the supremum is attained and equals $\frac{(L+\mu)^2}{4\mu L}$.

Remark. By homogeneity (both sides scale like $\|z\|_2^4$), the inequality is equivalent to $(z^\top Hz)(z^\top H^{-1}z) \le \frac{(L+\mu)^2}{4\mu L}\,\|z\|_2^4$ for all $z \neq 0$.

Question 30: *2 points.*

Consider the least-squares objective

$$F(\theta) = \frac{1}{2n} \|\Phi\theta - y\|_2^2, \qquad \Phi \in \mathbb{R}^{n\times d}, \quad y \in \mathbb{R}^n,$$

whose gradient is $\nabla F(\theta) = \frac{1}{n}\Phi^\top(\Phi\theta - y) = H(\theta - \eta_\star)$, where $H = \frac{1}{n}\Phi^\top\Phi \in \mathbb{R}^{d\times d}$ is the (constant) Hessian of F and $\eta_\star$ is any minimizer of F. Starting from θ_{t-1}, run one step of gradient descent with step size γ_t,

$$\theta_t = \theta_{t-1} - \gamma_t \nabla F(\theta_{t-1}).$$

Assume $\nabla F(\theta_{t-1}) \neq 0$. Show that the step size γ_t that minimizes $F(\theta_t)$ (exact line search) is

$$\gamma_t = \frac{\|\nabla F(\theta_{t-1})\|_2^2}{\nabla F(\theta_{t-1})^\top H \, \nabla F(\theta_{t-1})}.$$

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Solution: Write $g = \nabla F(\theta_{t-1})$ and $e = \theta_{t-1} - \eta_\star$, so that $g = He$. Since F is quadratic with Hessian H, an exact Taylor expansion around $\eta_\star$ (using $\nabla F(\eta_\star) = 0$) gives, for every θ,

$$F(\theta) - F(\eta_\star) = \tfrac{1}{2}(\theta - \eta_\star)^\top H(\theta - \eta_\star).$$

Hence the value after one step, as a function of γ, is

$$\varphi(\gamma) := F(\theta_{t-1} - \gamma g) - F(\eta_\star) = \tfrac{1}{2}(e - \gamma g)^\top H(e - \gamma g) = \tfrac{1}{2}e^\top He - \gamma\, g^\top He + \tfrac{\gamma^2}{2}\, g^\top Hg.$$

Because $He = g$, we have $g^\top He = g^\top g = \|g\|_2^2$, so

$$\varphi(\gamma) = \tfrac{1}{2}e^\top He - \gamma\|g\|_2^2 + \tfrac{\gamma^2}{2}\, g^\top Hg,$$

a convex quadratic in γ (since $g^\top Hg \geq 0$). Setting $\varphi'(\gamma) = -\|g\|_2^2 + \gamma\, g^\top Hg = 0$ yields

$$\gamma_t = \frac{\|g\|_2^2}{g^\top Hg} = \frac{\|\nabla F(\theta_{t-1})\|_2^2}{\nabla F(\theta_{t-1})^\top H \, \nabla F(\theta_{t-1})}.$$

The denominator is strictly positive whenever $g \neq 0$: indeed $g = He$ lies in the range of H, which is orthogonal to its kernel, so $g^\top Hg = 0$ would force $g = 0$.

Question 31: *3 points.*

Consider the same least-squares objective $F(\theta) = \frac{1}{2n}\|\Phi\theta - y\|_2^2$ as in the previous question, with constant Hessian $H = \frac{1}{n}\Phi^\top\Phi$ and minimizer $\eta_\star$, and run gradient descent $\theta_t = \theta_{t-1} - \gamma_t \nabla F(\theta_{t-1})$ with the exact-line-search step size $\gamma_t = \|\nabla F(\theta_{t-1})\|_2^2 / \big(\nabla F(\theta_{t-1})^\top H \nabla F(\theta_{t-1})\big)$ from the previous question. Assume $\nabla F(\theta_{t-1}) \neq 0$. Assume in addition that F is strongly convex, i.e. H is invertible; denote by $0 < \mu \leq L$ the smallest and largest eigenvalues of H and by $\kappa = L/\mu \geq 1$ the condition number.

Show that

$$F(\theta_t) - F(\eta_\star) \leq \left(\frac{\kappa - 1}{\kappa + 1}\right)^2 \big(F(\theta_{t-1}) - F(\eta_\star)\big),$$

and compare this rate with that of gradient descent with a constant step size.

You may use the Kantorovich inequality established earlier: $\sup_{\|z\|_2=1} (z^\top H z)(z^\top H^{-1} z) = \dfrac{(L+\mu)^2}{4\mu L}$.

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Solution: Keep the notation $g = \nabla F(\theta_{t-1}) = He$ and $e = \theta_{t-1} - \eta_\star$. From the previous question $\varphi(\gamma) = \frac{1}{2}e^\top He - \gamma\|g\|_2^2 + \frac{\gamma^2}{2} g^\top Hg$, and plugging in the minimizer $\gamma_t = \|g\|_2^2/(g^\top Hg)$ gives the optimal value

$$F(\theta_t) - F(\eta_\star) = \min_\gamma \varphi(\gamma) = \tfrac{1}{2}e^\top He - \frac{\|g\|_2^4}{2\, g^\top Hg}.$$

Since H is invertible, $e = H^{-1}g$, hence $e^\top He = g^\top H^{-1} g$ and $F(\theta_{t-1}) - F(\eta_\star) = \frac{1}{2}e^\top He = \frac{1}{2}g^\top H^{-1}g$. Therefore

$$\frac{F(\theta_t) - F(\eta_\star)}{F(\theta_{t-1}) - F(\eta_\star)} = \frac{g^\top H^{-1}g - \|g\|_2^4/(g^\top Hg)}{g^\top H^{-1} g} = 1 - \frac{\|g\|_2^4}{(g^\top Hg)(g^\top H^{-1}g)}.$$

Applying Kantorovich. By the Kantorovich inequality established earlier, in its homogeneous form $(z^\top Hz)(z^\top H^{-1}z) \leq \frac{(L+\mu)^2}{4\mu L}\|z\|_2^4$, applied to $z = g$,

$$(g^\top Hg)(g^\top H^{-1}g) \leq \frac{(L+\mu)^2}{4\mu L}\,\|g\|_2^4,$$

so $\frac{\|g\|_2^4}{(g^\top Hg)(g^\top H^{-1}g)} \geq \frac{4\mu L}{(L+\mu)^2}$ and

$$\frac{F(\theta_t) - F(\eta_\star)}{F(\theta_{t-1}) - F(\eta_\star)} \leq 1 - \frac{4\mu L}{(L+\mu)^2} = \frac{(L-\mu)^2}{(L+\mu)^2} = \left(\frac{\kappa-1}{\kappa+1}\right)^2,$$

the last equality using $\kappa = L/\mu$.

Comparison. The per-iteration factor $\left(\frac{\kappa-1}{\kappa+1}\right)^2$ is exactly the one achieved by constant-step gradient descent with the *optimal* step size $\gamma = \frac{2}{L+\mu}$, and it is smaller (better) than the factor $\left(1 - \frac{1}{\kappa}\right)^2 = \left(\frac{\kappa-1}{\kappa}\right)^2$ obtained with the simpler choice $\gamma = 1/L$ (since $\frac{\kappa-1}{\kappa+1} < \frac{\kappa-1}{\kappa}$). Thus, in the worst case, exact line search matches optimally-tuned constant-step gradient descent and improves on the conservative choice $\gamma = 1/L$.

Question 32: *1 points.*

In the previous two questions you analyzed two ways of choosing the step size for gradient descent on the least-squares objective F: the exact line search step size $\gamma_t = \arg\min_{\gamma \geq 0} F\big(\theta_{t-1} - \gamma \nabla F(\theta_{t-1})\big)$, and a constant step size (for instance $\gamma = 1/L$, or the optimal constant choice $\gamma = 2/(L+\mu)$).

Compare the two methods from a practical point of view. In particular, address:

- what each method needs to know about the problem (e.g. the constants μ and L); and
- what extra computation, if any, exact line search requires at each iteration.

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Solution: *Knowledge of the problem constants.* A constant step size is fixed in advance, and a good choice requires knowing the curvature constants: $\gamma = 1/L$ needs L, while the optimal $\gamma = 2/(L+\mu)$ needs both L and μ. If these are unknown or only loosely estimated, the step is either too small (needlessly slow) or too large (the iterates may fail to decrease F, or even diverge). Exact line search, by contrast, is *adaptive*: it requires no knowledge of μ or L. At each iteration it automatically picks the step that minimizes F along the current direction, so it never overshoots and needs no step-size tuning.
Extra computation. The price of this adaptivity is that every iteration must solve the auxiliary one-dimensional problem $\min_{\gamma \geq 0} F\big(\theta_{t-1} - \gamma \nabla F(\theta_{t-1})\big)$, whereas constant-step gradient descent simply plugs in a fixed number. For the quadratic least-squares objective this auxiliary problem has the closed form derived earlier, $\gamma_t = \|\nabla F(\theta_{t-1})\|_2^2 / \big(\nabla F(\theta_{t-1})^\top H \, \nabla F(\theta_{t-1})\big)$, costing only one extra matrix-vector product $H \, \nabla F(\theta_{t-1})$ per step, so the overhead is mild. For a general (non-quadratic) objective there is no closed form, and the line search becomes an inner optimization requiring several additional function/gradient evaluations at every outer step.
Summary. Exact line search trades the need to know and tune the constants μ and L for the cost of solving an auxiliary subproblem at each iteration. Since both methods share the same worst-case rate, line search is attractive precisely when μ and L are unknown and the auxiliary problem is cheap to solve (as in the quadratic case).

Question 33: *4 points.*

Let $F : \mathbb{R}^d \to \mathbb{R}$ be μ-strongly convex and L-smooth, with unique minimizer $\eta_\star$ and condition number $\kappa = L/\mu$. For $\theta \neq \eta_\star$, let α be the angle between the gradient $\nabla F(\theta)$ and the deviation to the optimum $\theta - \eta_\star$, defined by

$$\cos\alpha = \frac{\nabla F(\theta)^\top(\theta - \eta_\star)}{\|\nabla F(\theta)\|_2\, \|\theta - \eta_\star\|_2}$$

(equivalently, α is the angle between the descent direction $-\nabla F(\theta)$ and the direction $\eta_\star - \theta$ pointing to the optimum).

You may use the following two facts directly, without proving them.

- *Kantorovich inequality. (established earlier)* If $H \in \mathbb{R}^{d\times d}$ is symmetric positive definite with all its eigenvalues in $[\mu, L]$, then for every $z \neq 0$,

$$(z^\top H z)(z^\top H^{-1} z) \leq \frac{(L+\mu)^2}{4\mu L}\, \|z\|_2^4.$$

- *Co-coercivity.* If $G : \mathbb{R}^d \to \mathbb{R}$ is convex and β-smooth, then for all $\theta, \eta \in \mathbb{R}^d$,

$$\tfrac{1}{\beta}\, \|\nabla G(\theta) - \nabla G(\eta)\|_2^2 \leq \big(\nabla G(\theta) - \nabla G(\eta)\big)^\top (\theta - \eta).$$

(a) Suppose first that F is quadratic, so that $\nabla F(\theta) = H(\theta - \eta_\star)$ with constant Hessian $H = \nabla^2 F$, a symmetric positive-definite matrix whose eigenvalues lie in $[\mu, L]$. Show that

$$\cos\alpha \geq \frac{2\sqrt{\mu L}}{L+\mu}.$$

(b) Show that the same bound holds for an arbitrary (not necessarily quadratic) μ-strongly-convex, L-smooth function F. *Hint: apply co-coercivity to $G(\theta) = F(\theta) - \frac{\mu}{2}\|\theta\|_2^2$, which is convex and $(L-\mu)$-smooth.*

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Solution: **(a)** Write $e = \theta - \eta_\star$ and $g = \nabla F(\theta) = He$. Then

$$\cos\alpha = \frac{g^\top e}{\|g\|_2 \|e\|_2} = \frac{e^\top H e}{\sqrt{e^\top H^2 e}\,\sqrt{e^\top e}}.$$

Set $u = H^{1/2} e$ (well defined since $H \succ 0$); then $e^\top H e = \|u\|_2^2$, $e^\top e = u^\top H^{-1} u$ and $e^\top H^2 e = u^\top H u$, so

$$\cos\alpha = \frac{\|u\|_2^2}{\sqrt{(u^\top H u)(u^\top H^{-1} u)}}.$$

By the Kantorovich inequality $(u^\top H u)(u^\top H^{-1} u) \leq \frac{(L+\mu)^2}{4\mu L}\|u\|_2^4$, hence $\sqrt{(u^\top H u)(u^\top H^{-1} u)} \leq \frac{L+\mu}{2\sqrt{\mu L}}\|u\|_2^2$ and

$$\cos\alpha \geq \frac{\|u\|_2^2}{\frac{L+\mu}{2\sqrt{\mu L}}\|u\|_2^2} = \frac{2\sqrt{\mu L}}{L+\mu}.$$

(b) Let $G(\theta) = F(\theta) - \frac{\mu}{2}\|\theta\|_2^2$. Since F is μ-strongly convex, G is convex; since F is L-smooth, G is $(L-\mu)$-smooth, with gradient $\nabla G(\theta) = \nabla F(\theta) - \mu\theta$. Using $\nabla F(\eta_\star) = 0$ we get $\nabla G(\eta_\star) = -\mu\eta_\star$, so, writing $g = \nabla F(\theta)$ and $e = \theta - \eta_\star$,

$$\nabla G(\theta) - \nabla G(\eta_\star) = \nabla F(\theta) - \mu\theta + \mu\eta_\star = g - \mu e.$$

If $L = \mu$, then G is 0-smooth, so ∇G is constant. Hence $\nabla F(\theta) - \nabla F(\eta^\star) = \mu(\theta - \eta^\star)$, so $\nabla F(\theta) = \mu(\theta - \eta^\star)$, and therefore $\cos\alpha = 1$. This equals $2\sqrt{\mu L}/(L+\mu)$ when $L = \mu$. Co-coercivity applied at θ and $\eta_\star$, multiplied through by $\beta = L - \mu \geq 0$, reads

$$\|g - \mu e\|_2^2 \leq (L-\mu)\,(g - \mu e)^\top e.$$

Expanding both sides,

$$\|g\|_2^2 - 2\mu\, g^\top e + \mu^2 \|e\|_2^2 \le (L-\mu)\, g^\top e - (L-\mu)\mu\|e\|_2^2,$$

which rearranges to

$$\|g\|_2^2 - (L+\mu)\, g^\top e + L\mu\, \|e\|_2^2 \le 0, \qquad \text{i.e.} \qquad (L+\mu)\, g^\top e \ge \|g\|_2^2 + L\mu\, \|e\|_2^2.$$

By the Arithmetic Mean - Geometric Mean inequality $\|g\|_2^2 + L\mu\|e\|_2^2 \ge 2\sqrt{L\mu}\,\|g\|_2\|e\|_2$, so $(L+\mu)\, g^\top e \ge 2\sqrt{L\mu}\,\|g\|_2\|e\|_2$. Dividing by $(L+\mu)\|g\|_2\|e\|_2 > 0$ (here $g \neq 0$ because $\theta \neq \eta_\star$ and F is strongly convex),

$$\cos\alpha = \frac{g^\top e}{\|g\|_2\|e\|_2} \ge \frac{2\sqrt{\mu L}}{L+\mu}.$$